A1.2 Nucleic acids

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a1-2.html

Written for this site from the class slides. The chapter is on pages 14–24 of the course textbook.
Image credits are on the last slide of the slides for this subtopic.

Building blocks

A1.2.1 DNA is the genetic material of life

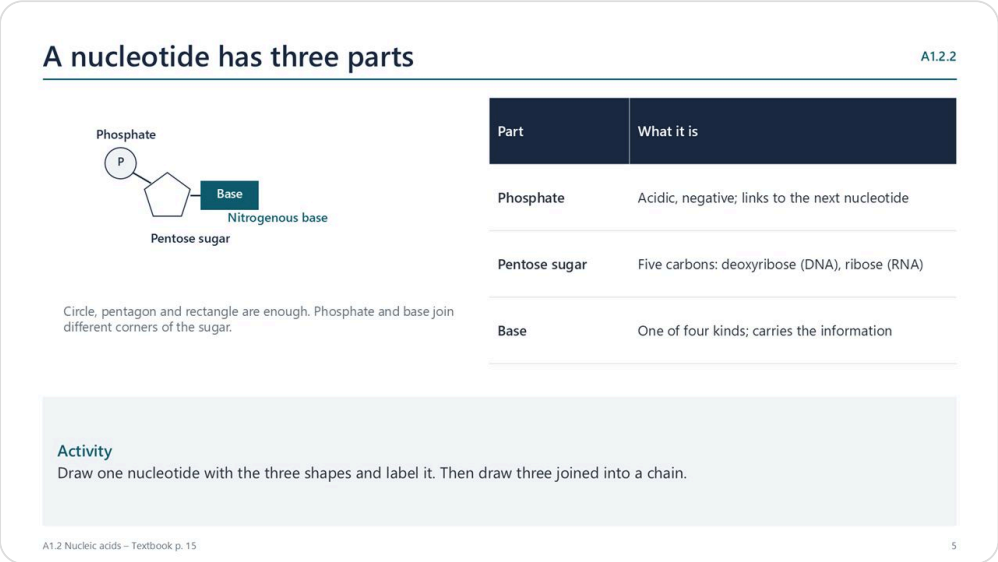
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All living organisms use DNA as their genetic material. Some viruses use RNA, but viruses are not considered to be living.

A1.2.2 The three parts of a nucleotide

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A nucleotide has a **phosphate**, a **pentose sugar** and a **nitrogenous base**. In diagrams, use a circle, a pentagon and a rectangle, in these positions.



A1.2.3 The sugar-phosphate backbone

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Each sugar bonds to the phosphate of the next nucleotide. This makes a continuous chain of covalently bonded atoms, so that the backbone is strong. Any base can follow any other.

A1.2.4 Bases carry the code

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DNA has **adenine, cytosine, guanine** and **thymine**. RNA has **uracil** in place of thymine. Only the base varies along a strand, so the base sequence carries the information.

A1.2.5 RNA is a polymer formed by condensation

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RNA nucleotides are monomers that join by **condensation**, releasing water for each new bond. RNA is a single strand with a sugar-phosphate backbone.

DNA

A1.2.6 The DNA double helix

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Two **antiparallel** strands, running in opposite directions, are linked by **hydrogen bonds** between complementary bases: A with T, and G with C.

Exam tip: draw the two strands antiparallel. You do not need to draw the helix, or to remember the numbers of hydrogen bonds.

A1.2.7 Differences between DNA and RNA

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	DNA	RNA
Strands	Two	One
Sugar	Deoxyribose	Ribose
Bases	A, C, G, T	A, C, G, U

Ribose has an -OH group on the carbon where deoxyribose has only -H, so deoxyribose has one oxygen atom fewer.

Using the structure

A1.2.8 Why base pairing matters

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Pairing depends on hydrogen bonds, which form only between the correct bases. Each strand can therefore act as a template in **replication**. In **expression**, DNA is a template for mRNA, and anticodons on tRNA pair with codons on mRNA.

A1.2.9 Limitless capacity to store information

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Any length of DNA molecule and any base sequence is possible. With four bases, there are 4 possible sequences for one position, 16 for two and 64 for three, and the number rises very quickly. So DNA stores data with great economy.

A1.2.10 The code as evidence of common ancestry

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Almost all organisms use the same genetic code. The chance of this arising separately many times is extremely small, so the best explanation is **universal common ancestry**.

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